

WildFire Simulator
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A. INTRODUCTION

During the training of new recruits for both wildfire and structural fire disciplines in British Columbia, there is no readily available software to help new recruits visualize how environmental variables, such as temperature, humidity, slope, fuel, or wind, affect forest fire behaviour. Having this would allow recruits to better understand fire behaviour and how to

contain it safely and effectively, before being asked to manage fires in person. This is important because new recruits in both wildfire and structural fire disciplines often do not experience the effects of these variables on wildfire behaviour until they are actively working at trying to contain a wildfire. The aim of this project is to create a proof of concept design which could lay the groundwork for the creation of a wildfire spread simulator to enhance the learning of these new recruits. This would allow new recruits to better understand these variables and improve their decision making and safety when working to manage a real forest fire.

B. BACKGROUND

Before assessing the information from my project, it is important to first review some basic wildfire mechanics, variables, ratings, and understand the use of cellular automata.

WILDFIRE MECHANICS OVERVIEW (British Columbia, 2026a):

- In order for a fire to occur we need 3 key factors:
 - 1. *Heat*:
 - This can come from an ignition source (such as lightning or a match), and can also take into account the temperature of the environment.
 - 2. *Oxygen*:
 - Continuous oxygen in the surrounding air allows the fire to burn more aggressively.
 - 3. *Fuel*:
 - These are things the fire can burn, such as grass, sticks, or trees.
 - The simulator created for this project uses 3 fuel types: grass, low tree, and tall trees to demonstrate 2 important concepts with fuel.
 - Grass and trees with low hanging limbs (i.e. "low trees") are highly combustible and burn quickly.
 - Trees which have been properly managed and do not have low hanging limbs (i.e. "tall trees") tend to not ignite quickly and, in some cases when they are providing shade, can slow down fire spread.
- These three factors (heat, oxygen, and fuel) make up what is known as the "fire triangle", as if you remove any one of these items, a fire will be unable to burn. If you increase any of these items with the triangle intact, the fire intensity will increase.
- Once the elements of the fire triangle are in place, we then have "multiplier" elements which can enhance or inhibit fire activity depending on their level of occurrence. These multipliers are:
 - 4. *Humidity*:
 - The amount of water in the air.
 - High humidity will reduce fire behaviour as it increases the moisture content in the fuels (wood, grass etc).
 - 5. *Slope*:
 - The angle of the terrain.

- Fire burns faster uphill for two main reasons:
 - 1. Preheating: The heat from a fire downslope will dry out fuels upslope, making them more combustible.
 - 2. Fuel Availability: The flame can reach more fuel as flame tends to reach upslope; allowing it to burn fuel beside and above it.
- These effects are exponentially increased as slope increases. A slope angle of 40 degrees or more is where the greatest slope effect is seen.
- 6. Wind:
 - Fire activity will be more aggressive downwind for three main reasons:
 - 1. Wind will push the fire into the fuel down wind, igniting the fuel at a more rapid pace.
 - 2. Wind delivers oxygen to the fire quickly, much like when you blow on a campfire to make it ignite.
 - 3. Wind helps to dry out fuel, making it more combustible.
 - As wind speed increases, these effects increase.

WILDFIRE DANGER RATINGS

In BC, wildfire risk is provided for each region of the province, which indicates what the current wildfire risk is. This is based on the environmental variables indicated above, as well as others not discussed in this project (these are discussed in Natural Resources Canada (2025a)). The danger ratings are (British Columbia, 2026b):

- **Low:** Fires may start and spread easily, but tend to self-extinguish.
- **Moderate:** Fuels will be drying, with an increased risk of fires starting and spreading.
- **High:** Fuels are very dry and new fires may start easily and burn aggressively.
- **Extreme:** Fuels are extremely dry and fires can start and spread very easily.

The range of values for the environmental variables for each of these danger ratings in Kelowna, BC are outlined in table 1 below. These are what were used to discretely organize the testing and data collection of this project, as these conditions in the simulator should match those expected in the real world.

TABLE 1: Typical weather for each fire danger rating in Kelowna, BC (Weather Spark, 2026)

Fire Danger Rating	Typical Season*	Temperature Range (°C)	Humidity Range (%)	Wind Range (km/h)
Low	Late Fall	<0 - 15°C	50-100%	0-60+
	Winter			
	Early Spring			
Moderate	Mid to late Spring	15-25°C	30-50%	0-60+
	Mid Fall			
High	Late Spring	25-30°C	20-30%	0-60+

	Summer			
	Early Fall			
Extreme	Summer	>30°C	<20%	0-60+

*There is variance and anomalies that occur year to year, this is a rough estimate based off of historical values for Kelowna, BC.

WILDFIRE RATE OF SPREAD:

- The rate of spread of a wildfire is simply a measure of how much distance is burnt by the fire over the amount of time it took to cover that distance. The range of wildfire spread for wildfires is typically between 0.1km/h and 8 km/h (Natural Resource Canada, 2025b), although there are anomalies that fall outside of this range (with rates of spread above 20km/h recorded in British Columbia in extreme conditions (British Columbia, 2026c)).

CELLULAR AUTOMATA OVERVIEW:

- Cellular automata is simply a means of modeling a system where: you have a grid of cells, each cell has a state, and each cell has a neighbourhood, which it effects, made up of its neighbouring cells (Nature of Code, 2024). This was chosen as it is a common technique used in forest fire modelling simulation (Zhou et al., 2025; Freire & DaCamara, 2018), and works well for the probabilistic model chosen.

C. SYSTEM ARCHITECTURE OVERVIEW

→ *ARCHITECTURE: The system architecture is divided into 4 layers:*

- **1. User Interface Layer**, which handles:
 - Parameter input from the user.
 - Fire ignition location.
 - Camera controls (i.e. rotating the terrain) from the user.
- **2. Grid System Layer**, which handles:
 - Creation of the 2D grid.
 - Mapping the terrain object to the cells of the 2D grid.
 - Collecting and storing slope and fuel load information in individual grid cells.
- **3. Simulation Mechanics Layer**, which handles:
 - Iterating through the burning cells, extinguishing those which have burned a given time increment, and igniting neighbouring cells that meet the ignition requirements.
 - The fire spread probability calculations to determine which neighbouring cells are ready to ignite.
- **4. Core Controller Layer**, which:
 - Acts as a central coordinator of the system.
 - Initializes all system components and handles the calling of updates to the appropriate layers during the update loop.

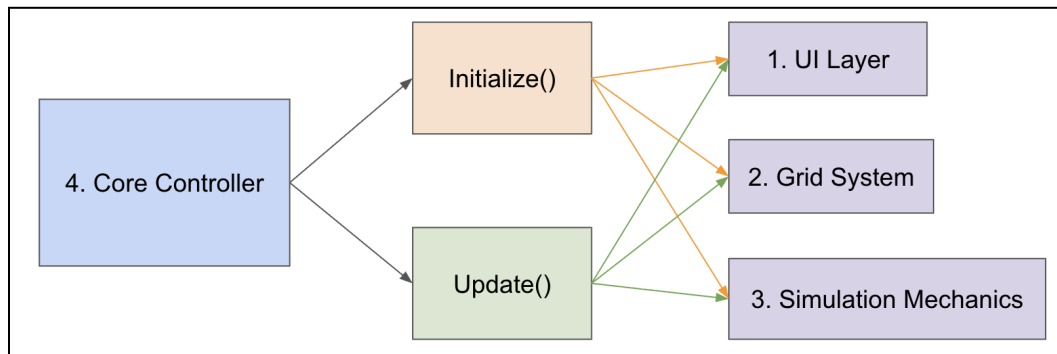


IMAGE 1: System Architecture Diagram

→ COMPONENTS: Each of the 4 layers of the system architecture has the following components:

- **1. User Interface Layer Components:**
 - *UI Manager*
 - Sets up event listeners for the data collection items (UI sliders).
 - Collects the user data once it is submitted.
 - Deactivates the data collection elements once it is submitted.
 - *Grid Interactor*
 - Responsible for the monitoring of the user's mouse over the terrain.
 - Detects and handles when the user clicks on a valid cell on the terrain.
 - *Camera Rotation*
 - Handles user input to make the terrain rotate to view it from different angles.
- **2. Grid System Components:**
 - *Grid Manager*
 - Responsible for creating the 2D backend grid, populating it with cell data, mapping it to the Unity Terrain object.
 - Handles the calculations of elevation and stores this in the grid cell.
- **3. Simulation Mechanics Components:**
 - *Burning Cell Manager*
 - Iterates through the burning cells, collects which ones are to be extinguished, and which neighbors need to be ignited.
 - Handles the igniting and extinguishing of cells.
 - *Ignition Probability Generator*
 - Handles the calculation of a neighbouring cell's probability of igniting based on the environment variables of the system.
- **4. Core Controller Components:**
 - *Fire Simulator Main*
 - Acts as a controller to ensure all components are initialized and updated appropriately.

→ DATA: For data storage, the program stores environmental and terrain information using data structures which represent grid cells of the 2D grid, and the user-defined parameters:

- **1. Grid Cell**
 - Will be placed at each cell of the 2D grid.
 - Stores:
 - Elevation
 - Fuel Load
 - Fuel load information is derived from the terrain objects within the cell.
 - Cell state (unburnt, extinguished or burning)
 - Cell location (x and z).
- **2. User Entry Data**
 - Used as an efficient means of transferring the parameters the user entered from the UI Layer to the Simulation Mechanics Layer.
 - Stores:
 - Wind speed and direction
 - Temperature
 - Humidity

→ INTERFACE: The user interface has the following format:

- **1. Environmental Parameters:** Collecting user parameters (wind speed and direction, temperature, humidity)
 - Sliders allow user entry within a fixed range.
 - Wind direction is a drop down selector of cardinal directions.
 - The start button sits below these and triggers the transfer of user data from UI to Simulation Mechanics.
 - A restart button used to reset the simulation as desired.
- **2. Cell Ignition:** Click to ignite:
 - This is only detected once parameters are submitted.
 - The system detects if the mouse was clicked.
 - It also checks if the click was on a valid grid cell.
- **3. Camera:** Camera Controls:
 - The user can press 'Q' or 'E' to rotate the terrain to the left or right respectively.
- **4. Compass:** Compass Display:
 - This is a small box that sits above the terrain to indicate which cardinal direction the user is viewing the terrain from.
 - This allows the user to remain oriented as they rotate the terrain.

→ SIMULATION EXAMPLE WORKFLOW:

1. 2D grid is created and populated on application load.
2. User sets environment parameters → clicks start.
 - a. Once started, environmental parameters cannot be changed.
3. User clicks on the terrain to ignite a cell.

4. Simulation update loop begins:
 - a. Burning cells evaluate their neighbours.
 - i. Ignition probabilities for only unburnt neighbours are calculated.
 - ii. Cells with suitable ignition probabilities are ignited.
 - b. Burning cells which have been burning for a set time frame will be extinguished.
 - c. Visualization updates cell colours (unburnt = green, burning = orange, extinguished = black).
 - d. Terrain is rotated whenever the user presses the buttons ('Q' or 'E') to do so.
-
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D. SIMULATION PROBABILITY MODEL

• 1. Grid Representation

- The Unity Terrain element is 150x150 Unity units, which was mapped to the 100x100 cells of the 2D backend grid. A mesh lay overtop of the Terrain element which was used to apply the colour changes.
 - I.e. each cell of the backend grid would appear as a 1.5x1.5 unit region on the terrain mesh.

• 2. Neighbour Ignition Model

- For the cellular automata program, a Moore's neighbourhood (8 neighbours) was used.
- For a given ignited cell, all 8 of its neighbours were individually inspected under the given environmental conditions to see if it meets the ignition requirements (with extinguished or burning neighbours ignored).
- Ignition of neighbours was initially far too exaggerated, and was restricted in 4 ways to create a more realistic ignition:
 - 1. Iterating through a cell's neighbours only on every 8th frame (instead of every frame). This was chosen through trial and error.
 - 2. Using weighted averaging of the environmental variables for each cell's probability calculation.
 - 3. Dampened the output of the overall probability for a given cell.
 - 4. Making the threshold of cell ignition be only if its probability is greater than twice a random double value (instead of a set threshold).
 - This also was added for slight variability as a threshold created slightly rigid and unrealistic ignition patterns.

• 3. Neighbour Ignition Calculation

$cellProbability = clamp01(base * (1 + wF * wW) * (1 + sF * sW) * (1 - hF * hW) * dampen)$

- **clamp01** → restrict the output to between 0 and 1
- **base** = (fuelFactor * fuelWeight) + (temperatureFactor * temperatureWeight)
- **wF, sF, hF** → wind, slope, and humidity factors, respectively
- **wW, sW, hW** → wind, slope, and humidity weights, respectively
- **dampen** → dampens the output as the fire activity was overall too extreme at all levels. This value was chosen through trial and error.

Formula Logic:

- **clamp01** → do not want the value returned to be negative or greater than 1.
 - **base** → this was chosen with regards to the "fire triangle" discussed above. A cell cannot ignite without its base elements: adequate fuel and temperature.
 - **wF, sF, hF, fF, tF** → the calculation of these factors are discussed below.
 - **wW, sW, hW, fW, tW** → these weights were chosen based on trial and error following the testing plan outlined below.
 - **Multiplying by: (1+wF*wW) and (1+sF*sW)** → slope and wind are items which can amplify fire behaviour and thus are multipliers of the base. The "1+" was added to both items so that if the wind or slope is 0, it does not nullify the entire formula.
 - **Multiplying by: (1-hF*hW)** → since humidity is a measure of moisture in the air, 1-humidity gives a level of dryness in the air, which can amplify fire behaviour.
- **4. Neighbour Ignition Environmental Variables Logic**
 - Fuel Factor Calculation
 - Fuel elements (trees and grass) are preset on the map.
 - During grid creation, each cell checks above it for any fuel objects and calculates its "fuelLoad" with this information.
 - $FuelFactor = clamp0Max(fuelLoad)$
 - clamp0Max → clamp the value between 0 and max allowed.
 - Max fuel load allowed chosen by trial and error based off of fuel load values received during map creation.
 - Temperature Factor Calculation
 - $TemperatureFactor = (temperatureChosen/maxTemperature)^2$
 - Calculation is squared as the relationship between fire and temperature is nonlinear (Gutierrez et al., 2021), and through testing this appeared more realistic.
 - Max temperature was chosen to display slightly above a peak Okanagan summer day.
 - Humidity Factor Calculation
 - $HumidityFactor = (humidityChosen/100)^2$
 - Calculation is squared as the relationship between fire and humidity is very heightened at the extremes. This produced more realistic effects of humidity in the simulator during testing.
 - Wind Factor Calculation
 - Wind factor was calculated as a dot product between wind direction and the direction of the neighbouring cell to the ignited parent cell.
 - Cardinal directions were represented as a 2D vector (normalized), as seen below:

NorthW (-0.7, 0.7)	North (0, 1)	NorthE (0.7, 0.7)
West (-1, 0)	Parent (0, 0)	East (1,0)

SouthW (-0.7, -0.7)	South (0, -1)	SouthE (0.7, -0.7)
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- $WindFactor = Max(0, dot(windDirection, neighbourDirection)) * (windSpeed)^2$
 - $Max(0, dot(...))$ → added as this means all neighbours not downwind from the parent will get a wind factor of 0.
 - Wind speed values were chosen to display the wind variance for the Okanagan region.
- Dot Product Table (normalized, wind direction: **NorthEast** = (0.7,0.7)):
 - Green and blue = positive wind factor (i.e. downwind)
 - Red = wind factor of 0 (i.e. upwind)

$(-0.7*0.7 + 0.7*0.7) = 0$	$(0*0.7 + 1*0.7) = 0.7$	$(0.7*0.7 + 0.7*0.7) = 1$
$(-1*0.7 + 0.7*0.7) = -0.7$	Parent (0,0)	$(1*0.7 + 0*0.7) = 0.7$
$(-0.7*0.7 + -0.7*0.7) = -1$	$(0*0.7 + -1*0.7) = -0.7$	$(0.7*0.7 + -0.7*0.7) = 0$

- Slope Factor Calculation
 - $SlopeFactor = (clamp01(slopeAngle/40))^2$
 - $clamp01$ → limit the value to between 0 and 1.
 - $slopeAngle$ → the angle of the slope of a line drawn between the parent and its neighbour.
 - $/40$ → 40 is used as this is the common cutoff for "steep slopes". Slopes greater than 40 tend to all have a maximal effect on fire activity (Wang et al., 2025).
 - The value is squared as the relationship between slope and fire activity is non-linear (Lee et al., 2009), and this produced realistic behaviour during testing.

E. IMPLEMENTATION

- **1. Software Development Model**
 - This project was built using an Incremental/ Agile approach.
 - The project was broken into manageable tasks.
 - Tasks were tested and integrated as they were completed.
- **2. Platform and Technology**
 - ENGINE:
 - Unity Game Engine was chosen as it has a built in 3D terrain system, allows for real-time rendering, and is widely used in industry for interactive 3D applications (Unity, 2026).
 - LANGUAGE:
 - The C# language was chosen as it is native to Unity software.

- **3. Project and Task Management**
 - GitHub was used as it allowed for the creation of Kanban boards for task management, could be linked directly to the Unity project using a local repository in GitHub Desktop, and is widely used for project management in software development (GitHub, 2026).
- **4. Version Control**
 - GitHub was used as it could track changes as they were made in my project, and managed clearly using local repositories on Github Desktop and remote repositories on the GitHub website.
- **5. Testing**
 - A structured testing plan was followed with each new change.
 - See the "G. Test Plan" section below.
- **6. Performance Considerations**
 - *1. Grid Size*
 - The grid was limited to 100x100 as this grid size allowed for clear visuals without performance limitations when tested.
 - *2. Terrain Mesh*
 - A mesh was added over top of the terrain to easily apply colour effects without altering the terrain object itself.
 - This colour change was completed only once per frame.
 - This replaced directly changing the colour of the terrain object which altered the terrain object in memory and needed to be manually reset after each simulation run.
 - *3. Cell Visuals*
 - Cells which are burning or extinguished are represented by a colour change and no particle physics were implemented to make realistic fire.
 - This was originally created then removed from the project as it noticeably impacted simulation responsiveness and was quite distracting overall.
 - *4. Cell Iteration*
 - Only cells which are burning are iterated.
 - Cells extinguish after a short time increment and are then removed from the list to limit the number of cells being iterated.
 - *5. Cell Ignition and Extinguishing*
 - Cell ignition was limited to every 8th frame to slow the rate of fire spread and improve overall simulation performance.
 - Cells are extinguished after 2 seconds to keep the iterating cells array small and display clear demarcation of burning and burnt regions on the map.
- **7. Environmental factors chosen**
 - Refer to the explanations provided in "E. Simulation Probability Model" above.
- **8. Project Scope**
 - This project was produced as a Computer Science Capstone Project, with the goal of creating a proof-of-concept design.

F. TESTING PLAN

• 1. Functional Unit Testing

- Unit tests were created and automated using Unity Testing Framework to test the accuracy of the ignition probability calculations.

• 2. Simulation Manual Testing

- Test conditions were based on the environmental conditions which make up the fire danger ratings used within British Columbia, outlined in Table 1 above.
- When a variable was changed (factor calculation or weighting), it would be tested in isolation with the base probability, under conditions outlined in table 2 below, until expected behaviour was achieved.
 - Once each variable was approved through the testing outlined in table 2 below, all variables were tested together in the simulator as outlined in table 3 below.
 - In doing this, it also enabled manual testing of the UI interface across the entire range of values, to ensure proper functioning.

TABLE 2: Testing After Parameter Change (gradually testing all items in isolation)

Temperature (°C)	Humidity (°C)	Wind Speed (km/h)	Wind Direction	Start Location	Slope
1. Base Probability (Temperature + Fuel)					
5°C	Not included (--)	--	--	SE Corner + within center each fuel type	--
20°C	--	--	--	SE Corner + within center each fuel type	--
40°C	--	--	--	SE Corner	--
2. Base Probability + Humidity (humidity focussed)					
5°C	90%	--	--	SE Corner + within center each fuel type	--
20°C	50%	--	--	SE Corner + within center	--

				each fuel type	
40°C	20%	--	--	SE Corner	--
3. Base Probability + Slope (slope focussed)					
5°C	--	--	--	SE Corner + base of steep slope	Tested on flat terrain + base of steep slope + center of steepest slope
20°C	--	--	--	SE Corner + base of steep slope	
40°C	--	--	--	SE Corner	
4. Base Probability + Wind (wind focussed) → Made sure wind is working in all directions					
20°C	--	10 km/h	N	SE Corner	--
20°C	--	20 km/h	S	NE Corner	--
20°C	--	40 km/h	E	SW Corner	--
20°C	--	60 km/h	W	SE Corner	--
5. Base Probability + Humidity + Slope					
5°C	90%	--	--	SE Corner + base of steep slope	Tested on flat terrain + base of steep slope + center of steepest slope
20°C	50%	--	--	SE Corner + base of steep slope	
40°C	20%	--	--	SE Corner	
6. Base Probability + Humidity + Wind → Testing at low and moderate humidity, as at high and extreme, wind effects become less isolated due to high ignition overall.					
5°C	90%	20 km/h	N	SE Corner + base of steep slope + in the center of all fuel types.	--
20°C	50%	20 km/h	N	SE Corner + base of steep	--

				slope + in the center of all fuel types.	--
5°C	90%	50 km/h	N	SE Corner + base of steep slope + in the center of all fuel types.	
20°C	50%	50 km/h	N	SE Corner (had adequate spread to assess all slope and fuel)	
7. All Variables (Base + Humidity + Wind + Slope) *This is table 3 below*					

TABLE 3: Testing Final Parameters**

Temperature (°C)	Humidity (%)	Wind Speed (km/h)	Wind Direction	Start location
Minimal Simulator Fire Conditions				
0°C	100%	0 km/h	N	SE Corner
Low Fire Danger (with and without wind)				
10°C	80%	0 km/h	N	SE Corner
10°C	80%	40 km/h	N	SE corner
Moderate Fire Danger (with and without wind)				
20°C	50%	0 km/h	N	SE Corner
20°C	50%	40 km/h	N	SE Corner
High Fire Danger (with and without wind)				
35°C	30%	0 km/h	N	SE Corner
35°C	30%	40 km/h	N	SE Corner
Extreme Fire Danger (with and without wind)				

40°C	20%	0 km/h	N	SE Corner
40°C	20%	40 km/h	N	SE Corner
Maximal Simulator Fire Conditions				
50°C	0%	60 km/h	N	SE Corner

**The overall outcome of each condition is discussed in the Results section below

G. RESULTS

Reading the user manual (Appendix A (page 26)) at this point will help with understanding the different fuel types and colours on the terrain images used in this section, if needed.

The results for this project were obtained from testing the conditions outlined in Table 3 above. Each condition was tested a minimum of 4 times and an average time and total cells burned were gathered after each run. This is shown in Table 5 below. Each of the trials was completed in an effort to see the fire behaviour on both flat and sloped ground, as well as within all of the fuel types. Given that some parameter conditions lacked adequate fire spread, trials were modified (as per table 4 below) to ensure these variables (slope and fuel) were measured in each condition. This is also displayed visually in table 6 below.

Table 4: Structure of how each of the 4 trials were conducted for observing each condition

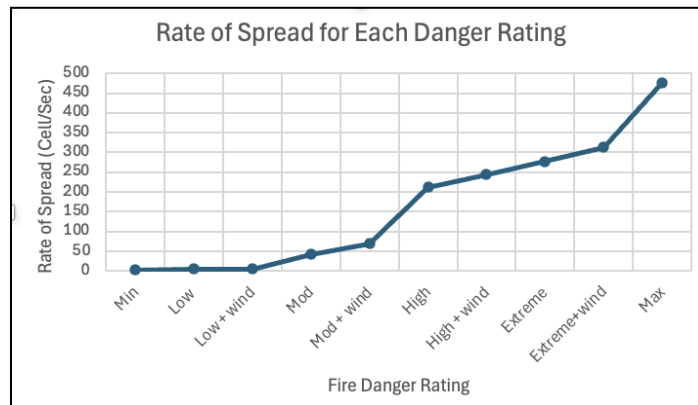
Trial Number	Fuel + Slope	
	If trial 1 ignited poorly	If trial 1 ignited well
1	Regular fuel + Flat ground	--
2	Grass + Flat Ground and Grass + Steep slope	Regular Fuel + Flat ground
3	Low trees + Base of slope	Regular Fuel + Flat ground
4	Tall Trees + Base of slope	Regular Fuel + Flat ground

Table 5: Data and Rate of Spread Calculations From Each Tested Condition

Condition	Time to Extinguish (seconds)				Total Cells Burned				Rate of Spread (Cells/second)	Percentage of the map burned (%)**
	Trials				Trials					
	1	2	3	4	1	2	3	4		
Minimal Simulator	9	7	11	9	41	13	48	25	3.44	0.31%

Fire Conditions	Avg: 9 sec				Avg: 31 cells					
Low Fire Danger (no wind)	14s	8s	12s	10s	85	62	35	57	5.23	0.58%
	Avg: 11 sec				Avg: 57.5 cells					
Low Fire Danger (with wind)	11s	18s	15s	12s	55	120	85	70	5.89	0.83%
	Avg: 14 sec				Avg: 82.5 cells					
Moderate Fire Danger (no wind)	51 s	1m 40 s	2m 15s	47s	672	6313	6810	487	42.51	35.71%
	Avg: 1min 24sec				Avg: 3,571 cells					
Moderate Fire Danger (with wind)	1m 42s	2m 10s	2m 15s	1m 55s	8533	8412	8587	8071	69.43	84.01%
	Avg: 2 min 1 sec				Avg: 8,401 cells					
High Fire Danger (no wind)	44s	49s	48s	45s	9987	9992	9975	9984	212.45	99.85%
	Avg: 47 sec				Avg: 9,985 cells					
High Fire Danger (with wind)	39s	45s	40s	40s	9993	9991	9995	9991	243.73	99.93%
	Avg: 41 sec				Avg: 9,993 cells					
Extreme Fire Danger (no wind)	36s	38s	35s	36s	9998	9998	9997	9998	277.72	99.98%
	Avg: 36 sec				Avg: 9,998 cells					
Extreme Fire Danger (with wind)	31s	33s	31s	32s	9999	9998	10000	9998	312.44	99.98%
	Avg: 32 sec				Avg: 9,998 cells					
Maximal Simulator Fire Condition	20s	21s	21s	20s	10000	10000	10000	10000	476.19	100.00%
	Avg: 21 sec				Avg: 10,000 cells					

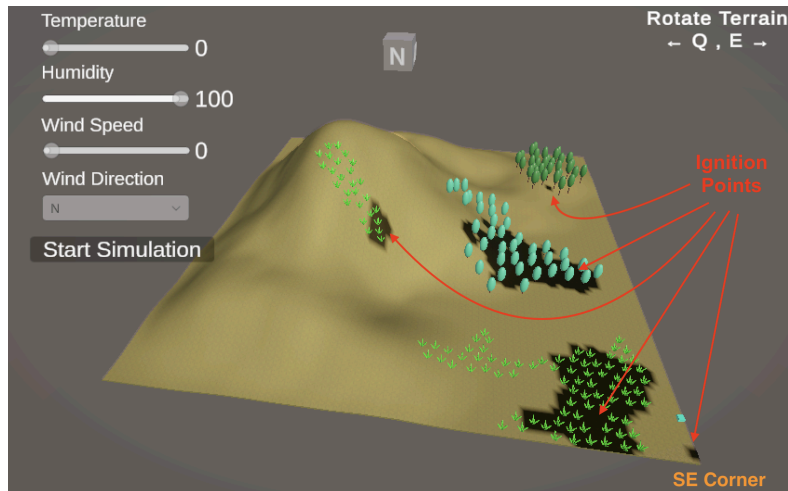
***Percentage burned was calculated by taking the number of cells burned in the trial (printed out in the code) and then dividing this by 10,000 (since there are 10,000 cells in the 100x100 grid).*



GRAPH 1: Rate of spread mapped against the corresponding danger ratings outlined in Table 5 above.

TABLE 6: Visual Display and Outline of the Final Parameter Results

Temperature (°C)	Humidity (%)	Wind Speed (km/h)	Wind Direction	Start location	Slope	Fuel Activity
Minimal Simulator Fire Conditions						
0°C	100%	0 km/h	N	SE Corner Ignition	Minimal to no effect on fire spread.	Fire spreads lightly through grass and low trees. No fire spread in high trees.
Average time to fully extinguish: 9 seconds Overall Outcome: - Minimal to no fire spread overall. - Fire does not spread in areas of low fuel. - In areas of adequate fuel fire spread increases but remains minimal and patchy.						
IMAGE 2: Minimal simulator fire conditions (looking North) - Visible small burn plots within the heavily fuelled regions. These are all individual fires which were ignited within the fuelled regions. - Fire spread in areas of lower fuel do not burn (top right and bottom right).						



Low Fire Danger (without wind)

10°C	80%	0 km/h	N	SE Corner Ignition	Slope effects are only noticeable where there is adequate fuel.	Fuel burns well but not fully though.
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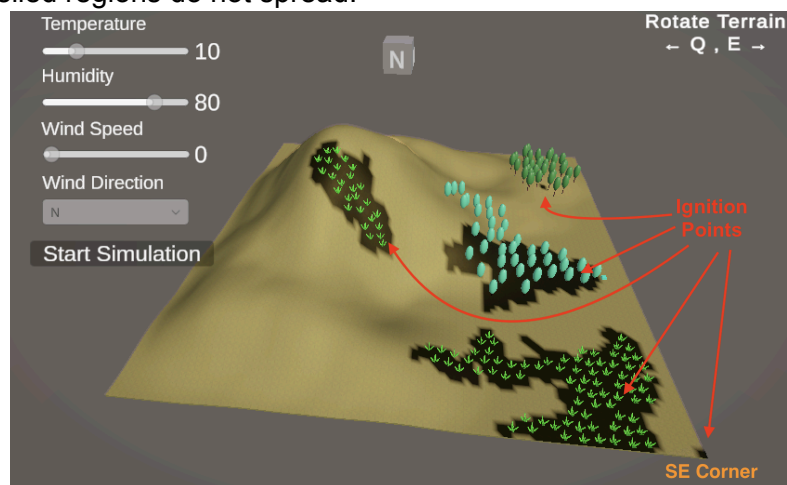
Average time to fully extinguish: 11 seconds

Overall Outcome:

- Fire spread was slightly increased but isolated to areas of higher fuel load only.
 - Fire spread in areas of lower fuel was minimal to none. Only in regions carrying more energy from burning fuel.
 - The burning of fuelled regions remains slightly patchy.

IMAGE 3: Low fire danger without wind (looking North)

- Fuelled regions have burned more thoroughly but not fully in some regions.
- Regions which are sloped and fuelled burn thoroughly.
 - Each fuelled region was an individual ignition.
- Lower fuelled regions do not spread.



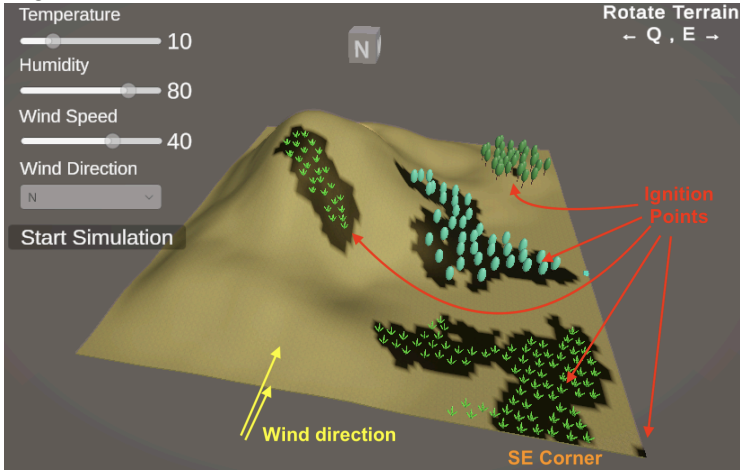
Low Fire Danger (with wind)						
10°C	80%	40 km/h	N	SE Corner Ignition	Slope effects are only noticeable where there is adequate fuel.	Fuel burns well and thoroughly if it is predominantly down wind.
Average time to fully extinguish: 14 seconds Overall Outcome: - Fire spread was slightly increased but isolated to areas of higher fuel load only. - Fire spread in areas of lower fuel was minimal to none. Only in regions carrying more energy from burning fuel. - The burning of fuelled regions remains slightly patchy, unless they are downwind, where the fuel burns more thoroughly.						
IMAGE 4: Low fire danger with wind (looking North) - Similar burn patterns and behaviour to without wind in the same conditions. - Noticeable slight increase in fire spread on the downwind (north) side of all burn plots.						
						
Moderate Fire Danger (without wind)						
20°C	50%	0 km/h	N	SE Corner Ignition	Slope effects are much more noticeable.	Fuel ignition is increased, especially when combined with slope.
Average time to fully extinguish: 1 minute 24 seconds Overall Outcome: - Fire burned a much larger region, fuel burned well and slope effects became more apparent. The burn region remained spotty and rate of spread was moderate to minimal. - One thing to note is that in some trials the ignition point did not spread right away. Some trials took 2-3 ignitions for the fire to begin spreading, which does match real world conditions for this. - This setting seemed to have the most variability between trials.						

IMAGE 5: Moderate Fire Danger without wind, Trial 2 (looking North)

- Much larger burn plot from a single ignition point.
- Burnt region is spotty and did not burn the edges of the map thoroughly.

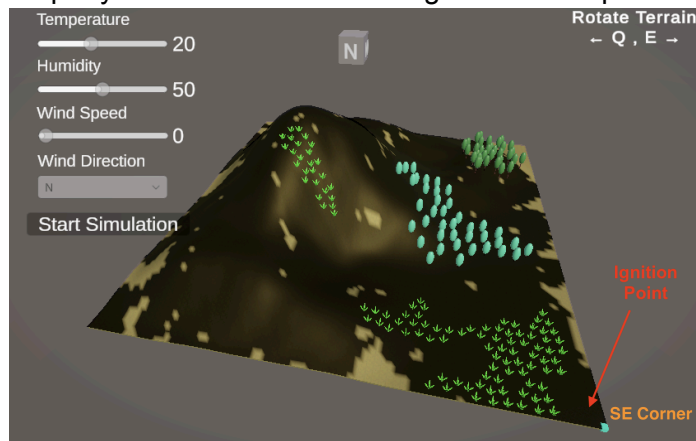


IMAGE 6: Moderate Fire Danger without wind, Trial 4 (looking North)

- This stage had the most variance in fire ignition patterns, as seen in this image from trial 4 in the same conditions.

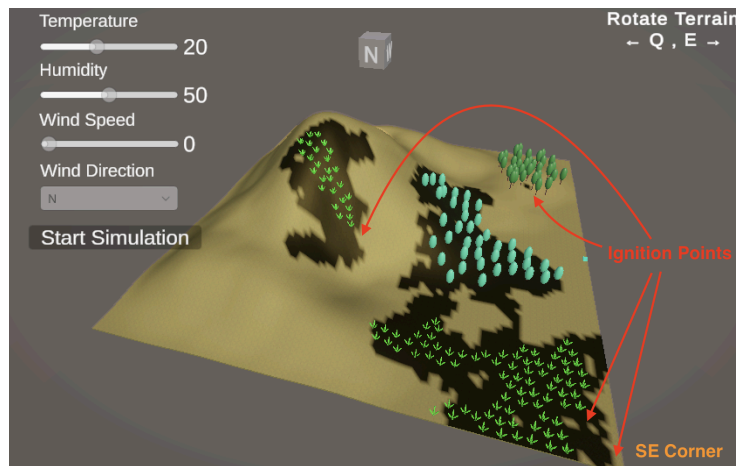
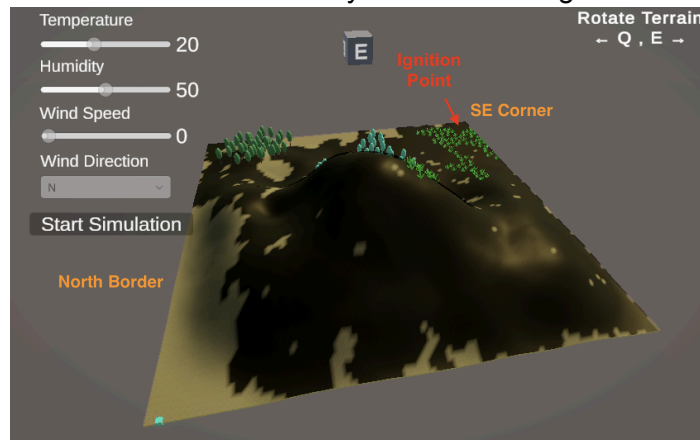


IMAGE 7: Moderate Fire Danger without wind, Trial 2 (looking East)

- Burnt region does not burn continuously to the north edge of the map.



Moderate Fire Danger (with wind)						
20°C	50%	40 km/h	N	SE Corner Ignition	Slope effects are much more noticeable.	Fuel ignition is increased, especially when combined with slope and the added wind.

Average time to fully extinguish: 2 minutes 1 second

Overall Outcome:

- Similar burn patterns behaviours to the same conditions without wind except down wind regions burnt noticeably more with the added wind.
- The burn plot was burnt slightly more thorough than without wind in these conditions.

IMAGE 8: Moderate fire danger with wind (looking north)

- Similar burn pattern to without wind in the same condition, although regions north of the ignition point burned slightly more thoroughly.

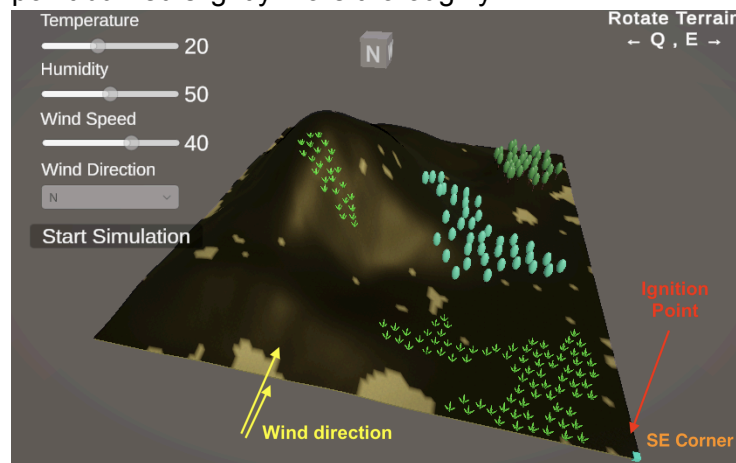
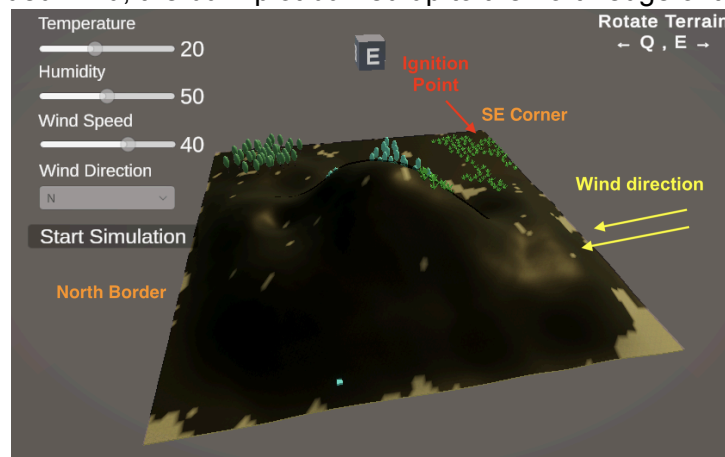


IMAGE 9: Moderate fire danger with wind (looking east)

- With the added wind, the burn plot burned up to the north edge of the map.



High Fire Danger (without wind)						
35°C	30%	0 km/h	N	SE	Slope effects are	Fuel remains very

				Corner Ignition	more noticeable than at moderate level.	flammable, more so than in moderate conditions.
--	--	--	--	-----------------	---	---

Average time to fully extinguish: 47 seconds

Overall Outcome:

- Notable increase in fire rate of spread, as seen in the time to extinguish compared to the moderate conditions.
- The fire burnt the region much more thoroughly, with very little fuel left unburnt.

IMAGE 10: High Fire Danger without wind (looking north)

- The burnt region is very thorough with the entire map burn from a single ignition point.

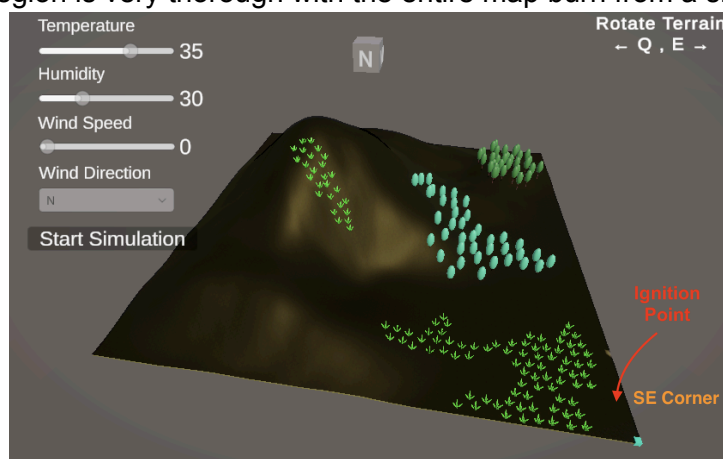
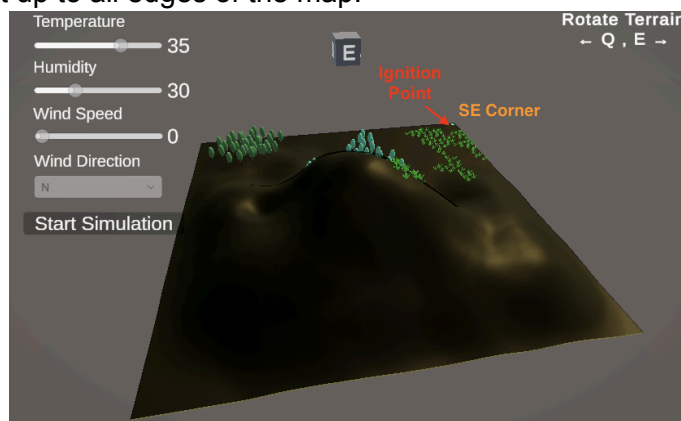


IMAGE 11: High Fire Danger without wind (looking east)

- The fire burnt up to all edges of the map.



High Fire Danger (with wind)

35°C	30%	40 km/h	N	SE Corner Ignition	Slope effects are more noticeable than at moderate level.	Fuel remains very flammable, more so than in moderate conditions. Grass and low tree rate of spread has a noticeable increase,
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						while tall trees have a noticeable decrease in rate of spread.
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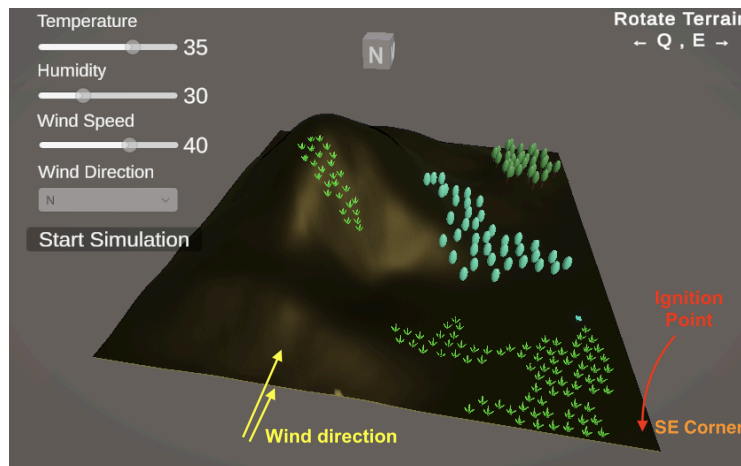
Average time to fully extinguish: 41 seconds

Overall Outcome:

- Very similar behaviour to the same conditions without wind, just slight increase in the rate of spread with the added wind effect.

IMAGE 12: High Fire Danger with wind (looking north)

- Much like these conditions without wind, the fire burnt the region very thoroughly from a single ignition.



Extreme Fire Danger (without wind)

40°C	20%	0 km/h	N	SE Corner Ignition	Slope effects are significant.	All fuel burns thoroughly, noticeable increase in burn rate in grass and low trees, tall tree rate of spread is slightly reduced but minimal.
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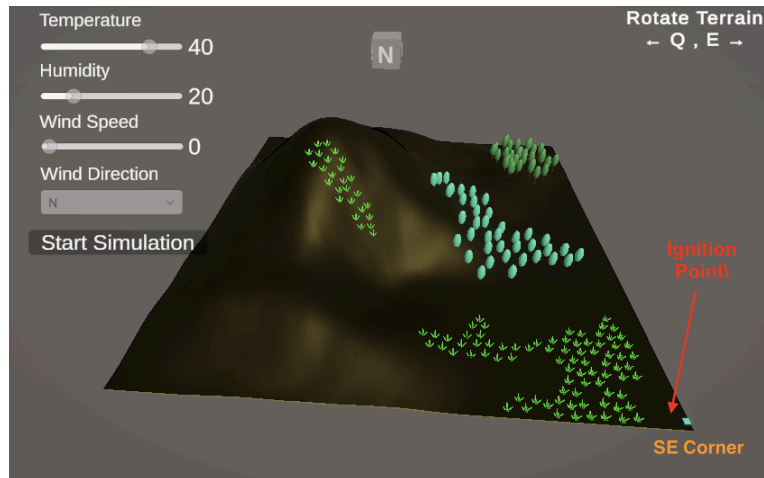
Average time to fully extinguish: 36 seconds

Overall Outcome:

- The fire burns the region quickly and thoroughly with very little hesitation in the rate of spread.

IMAGE 13: Extreme Fire Danger without wind (looking north)

- The fire burnt the region very thoroughly from a single ignition, leaving nothing unburnt.



Extreme Fire Danger (with wind)

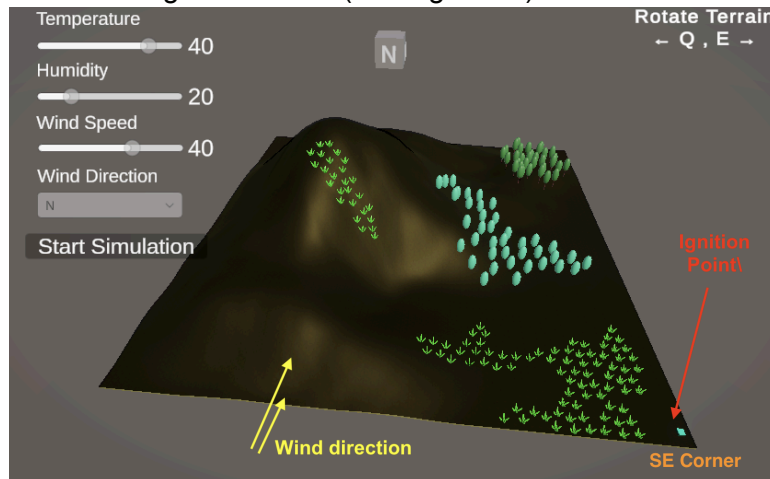
40°C	20%	40 km/h	N	SE Corner Ignition	Same as the same conditions above without wind.	Same as the same conditions above without wind.
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Average time to fully extinguish: 32 seconds

Overall Outcome:

- Same result as these conditions without wind above, but a slight increase in rate of spread with the added wind effect.

IMAGE 14: Extreme Fire Danger with wind (looking North)



Maximal Simulator Fire Conditions

50°C	0%	60 km/h	N	SE Corner Ignition	Since the rate of spread is so fast, changes in slope are not as noticeable.	Since the rate of spread is so fast, changes in fuel are not as noticeable.
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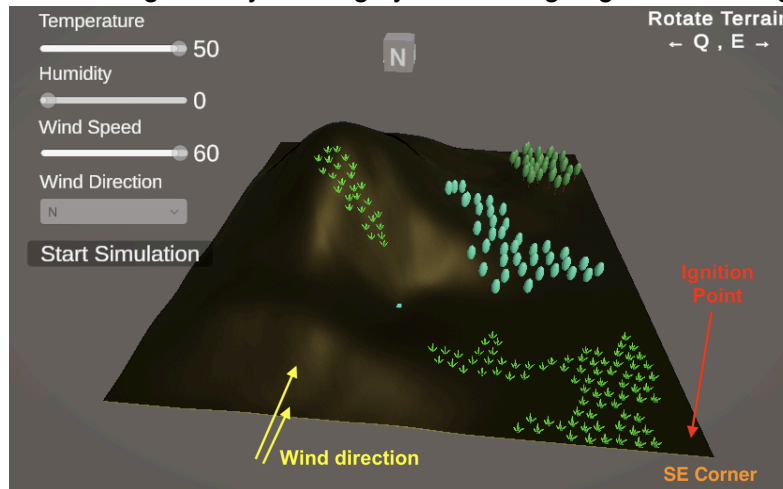
Average time to fully extinguish: 21 seconds

Overall Outcome:

- The fire burns incredibly fast and thoroughly throughout the terrain, travelling at nearly the same rate in all portions of the terrain.

IMAGE 15: Maximal Simulator Fire Conditions (looking north)

- The fire burnt the region very thoroughly from a single ignition, leaving nothing unburnt.



H. RESULTS DISCUSSION

1. Reviewing the results

The fire behaviour emulated by the simulator is reminiscent of that seen in actual fire behaviour. The trends seen in tables 5 above clearly displays that as we simulate the conditions of ever-increasing fire danger ratings, our fire rate of spread increases from 5.23 cells per second at a low fire danger rating to 312.44 cells per second at an extreme fire danger rating. This is also displayed in the total map area that is burnt, which increases from 0.58% at a low fire danger rating to 99.98% at an extreme fire danger rating.

Within these trends, the observations from each trial demonstrate how the effects of the environmental parameters played a role in the fire behaviour seen (in table 6 above):

- **TEMPERATURE AND HUMIDITY:**
 - In real world scenarios, these factors are huge drivers in determining and predicting fire behaviour. A key term relating these two parameters is known as "cross-over" which means when the humidity value (%) becomes less than the temperature value (C). When this occurs, a large spike in wildfire activity ensues. This is seen in the simulator as well, where through the low and moderate fire danger simulations, the fire behaviour gradually begins to increase, although ignition is still inconsistent at the moderate level. Once the high fire danger rating is simulated, we have the crossover effect, and the fire activity increases significantly, as shown in Graph 1 in the Results section above.
- **SLOPE:**

- At lower danger ratings, the effects of slope were not overly prominent unless there was adequate fuel to ignite.
- Once simulating moderate intensity and above was reached, the rate of spread moving up slope increased to a noticeable level, and became more pronounced at the high and extreme danger rating levels.
- **FUEL**
 - Fuel is a primary driver for all fire activity. Much like in the real world, if there is adequate dry fuel, a fire will spread in most conditions. The extent of this spreading depends on the quantity of fuel. Even at the lowest danger rating levels, the fire still spread lightly in heavily fueled regions, but did not spread beyond these regions until a moderate or higher danger rating condition level, matching the expected real world behaviour.
- **WIND**
 - At each simulated fire danger rating, there was a simulation with and without wind. As seen in Graph 1 above, the fire rate of spread increases in each iteration with the addition of wind. This was as expected, given the tested simulations were started at the south end of the simulation terrain, while the wind was going north, allowing ample fuel availability down wind.

2. Comparing results to expected fire behaviour

- To assess the results of the simulator to more tangible real world parameters, we can compare them to the expected fire behaviour as per the definitions and parameters laid out for each of the fire danger rating levels (IAWF, 2022). This is outlined in table 7 below. As we can see in this table, the expected behaviour and simulator behaviour match nicely, and the rates of spread, when adjusted for the simulator speed, display a similar rate of spread pattern to what would be expected in the real world.

Table 7: Expected versus simulator fire behaviour and rate of spread

Fire Danger Rating	Expected fire behaviour	Simulator Fire Behaviour	Expected Rate of Spread	Simulator Rate of Spread (km/h)*	Simulator Adjusted Rate of Spread (km/h)**
Low	- Fires difficult to ignite. - Fires unlikely to spread. - Fires likely to self extinguish.	- Fire spread is minimal to none. - Fire self extinguishes.	0 - 0.05 km/h	28 - 32 km/h	0.04 - 0.06 km/h
Moderate	- Fires spread slowly. - Fires may begin to be self-sustaining.	- Fire spreads slowly and sometimes self contains and spreads further.	0.03 - 0.5 km/h	226 - 375 km/h	0.3 - 0.7 km/h

High	- Fires spread becomes more active and rapid.	- Fire spread spreads the entire map and burns nearly every cell.	0.1 - 2+ km/h	1147 - 1312 km/h	1.9 - 2.2 km/h
Extreme	- Fires spread very rapidly.	- Fire spreads very aggressively.	1 - 4+ km/h	1495 - 1685 km/h	2.4 - 2.8 km/h

*Simulator rate of spread was converted into km/h by converting each cell unit to be equivalent to 1m. A cell which is 1.5unitsx1.5units is 1.5 meters x 1.5 meters. Formula for calculation is:
Rate of spread (km/h) = $\frac{\text{Cells}}{\text{second}} \times \frac{1.5 \text{ meters}}{\text{cell}} \times \frac{1 \text{ km}}{1000 \text{ meters}} \times \frac{60 \text{ seconds}}{1 \text{ minute}} \times \frac{60 \text{ minutes}}{\text{hour}}$. A cell size of 1.5 meters was chosen as the simulation animations were built on this assumption.

**Adjusted rate of spread is simply: $\frac{\text{Cells}}{\text{second}} \div 600$. This was determined by plotting the expected and simulator rates of spread and adjusting by the determined margin. This is outlined in Appendix C: Adjusting Simulator Rate of Spread.

With this data we can even go one step further to get a rough estimate of the real-world burning time and size. Since the ratio used in the table above to adjust the simulator rate of spread to the real world rate of spread is 600, we can apply this same ratio to the simulator time to get a time scale estimate. For example, if we take the average burn time for the "Moderate Fire Danger with wind", which was 2 minutes and 1 second, and scale this by 600, this would be a time of 20 hours and 10 minutes to burn 12.6km² (as it burned 8,501 cells and cells are 1.5m²). This provides a rough estimate and demonstrates how this could be applied to the simulator for future improvements. These estimates are represented in table 8 below. When comparing these values to actual fire data (British Columbia, 2026c), the values are within a realistic range.

Table 8: Estimated real world times and sizes for the simulated fires.

Fire Danger Rating	Average Simulator Burn Time (seconds)	Average Simulator Burn Size (number of cells)	Estimated Real-world Time (hours)	Estimate Real-world Size (kilometers ²)
Low	11 - 14 s	57 - 83	2 - 2.5 h	0.1 - 0.2 km ²
Moderate	84 - 121 s	3,570 - 8,401	14 - 20 h	5 - 13 km ²
High	41 - 47 s	9,985 - 9,993	6 - 8 h	14 - 15 km ²
Extreme	32 - 36 s	9,998	5 - 6 h	15 km ²

I. LIMITATIONS

Although this project provides a working wildfire spread probability simulation, it does have its limitations:

1. It does not demonstrate fire spotting activity ahead of the head of the fire.
 2. Wind speed ranges were kept at a smaller range for a more efficient testing purpose. Wind speeds can reach higher than the simulator range of 0-60 km/h.
 3. The simulator does not account for sun facing slopes being drier and more flammable, resulting in fires being more active on these slopes.
 4. The terrain and topography is preset and cannot be changed by the user to represent a specific landscape.
 5. The UI elements could be more polished for a production level simulator.
 6. The simulator currently can only be run within the Unity Game Engine, by someone who knows how to use this software. This would be limiting for most users.
 7. The fire activity is mapped as a static environment and has no time estimate with the simulation, so although it maps activity, it portrays no concept of timelines for this occurring.
 8. Along with the point above, the weather and variables (humidity, temperature etc) change throughout the day, which is not represented in the simulation static setting of the variables.
 9. The program can only be run on a desktop or laptop computer at present.
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J. FUTURE IMPROVEMENTS

There are several improvements that could be made to this project to enhance it from a proof of concept model to a production level simulator:

1. Creating the simulator to map specific terrains of interest for a given region. An alternative would be for the simulator to work with a topographic mapping software and be applied to any region.
2. The addition of fire physics could enhance the user experience.
3. Sound effects could be added for added engagement of the user.
4. The application could be published to a user-friendly application or web-server host for ease of access and use.
5. Adding a pause button for the user to pause the simulation as desired.
6. Implement a larger neighbourhood per cell to imitate fire spotting behaviour.
7. Increase the range of wind speeds or add variable wind behaviour for more realistic modeling.
8. Allow the simulator to map fire activity over a given day with changing temperatures, humidity values etc., to allow the simulation to appear more realistic.
9. Allow the simulation to account for sun facing slopes, as fire activity is more aggressive on sun facing slopes.
10. Allow the simulator to show timelines with the fire burning to map the fire behaviour over a given time interval.
 - a. This is outlined briefly as to how it could be done in the Results Discussion above.
11. Modify the simulator to run on mobile devices, tablets or VR/AR simulators.

12. A key item of fire spread is also how deep the fire burns into the ground, the cells of the grid could hold a "moisture" value which determines how deep the fire burns into the ground as well.
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K. CONCLUSION

In this project, a forest fire simulator using weighted probabilities within a cellular automata framework was built for use as an educational tool for wildfire and structural fire new recruits. The simulator presented a means of displaying the changes in fire activity as temperature, humidity, wind speed and direction, slope, and fuel are adjusted in a given map environment.

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M. APPENDICES

a. APPENDIX A: USER MANUAL

- i. When the simulator first opens, you will see (image 15 below):
 1. The terrain in the center of the screen,
 2. Toggles and a drop down in the top left to select temperature, humidity, and wind values.
 3. A compass square directly above the terrain (it will be a box with an "N" on it).
 - a. The cardinal direction this is showing is the direction you are facing. If the compass says "N" you are facing north.
 4. You can rotate the terrain to the left or right using the "Q" and "E" keys on your keyboard. This is indicated in the top right of the screen.

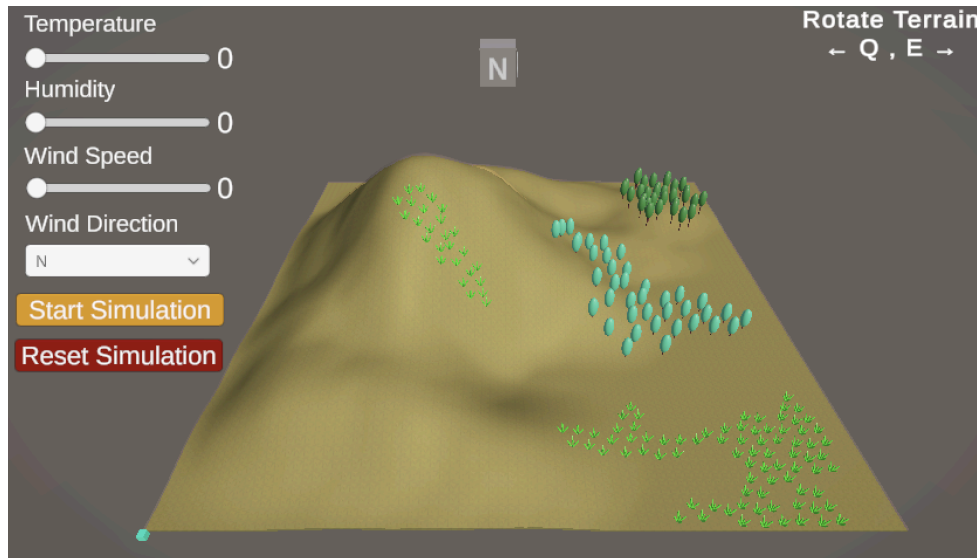


IMAGE 15: *The simulator opening scene*

ii. Running the Simulation:

1. Click and drag the toggles for temperature, humidity, and wind speed to select your desired value.
2. Use the drop down to select the wind direction.
3. Click start
 - a. You will not be able to start a fire until you press start.
 - b. Parameters cannot be changed once you press start.
4. Click a spot on the terrain where you would like the fire to ignite. The fire will start.
5. Rotate the terrain to watch the fire spread.
6. You can ignite multiple fires if desired.
7. Press the "Reset" button to restart the simulation.

iii. Terrain Explanation (Image 16 below):

1. There will be a blue box under your cursor showing you where on the map you are going to select.
2. For vegetation there is grass, high trees and low trees.
 - a. Fire burns faster in the grass.
 - b. High trees = trees with no ladder fuel, so fire should slow down in these areas.
 - c. Low trees = trees with ladder fuel, so fire will burn faster through these.
3. Burning ground is orange and extinguished ground is black.

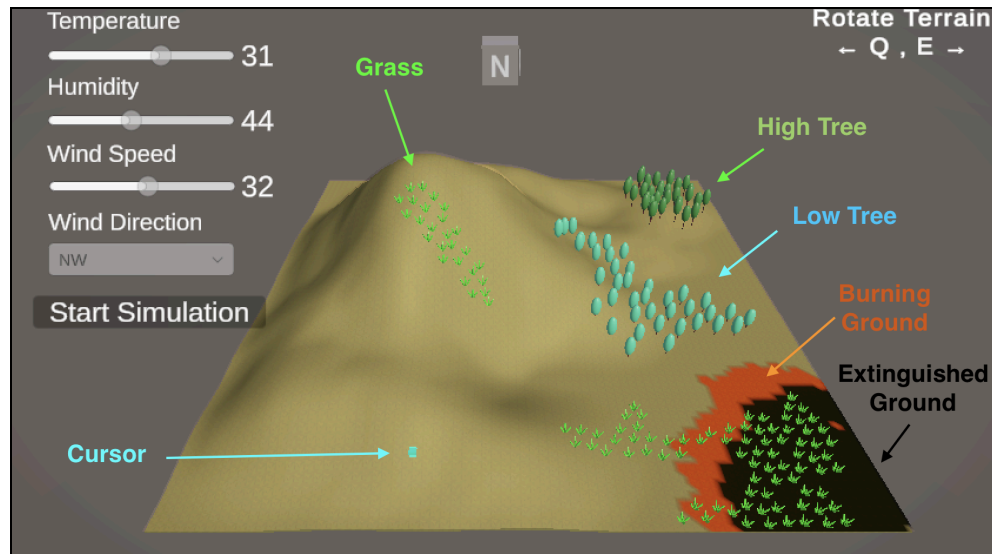


IMAGE 16: The simulator when running

- iv. Fire Behaviour in the simulator:
 1. The simulator considers the following variables:
 - a. Fuel
 - i. Grass (burns quickly), low trees (burn quickly), high trees (burn slowly).
 - b. Slope
 - i. Fire burns faster upslope.
 - c. Wind
 - i. Fire burns faster downwind, relative to wind speed.
 - d. Temperature:
 - i. Fire burns faster in higher temperatures.
 - e. Humidity:
 - i. Fire burns faster in lower humidities.
 2. You can adjust the parameters and where the fire is started to see these variables in action.

b. APPENDIX B: README

- i. (Katija, 2022) ← used to understand markdown formatting

Forest Fire Simulator:

A Unity-based simulation that models forest fire spread across a grid terrain using probabilistic ignition influenced by wind, fuel, temperature, humidity, and slope parameters.

Features:

- Grid-based fire propagation model
- Adjustable environmental parameters (wind, humidity, temperature)

- Real-time visualization of fire spread
- Interactive ignition point selection
- Unit testing completed using Unity Testing Framework

Running the project:

1. Open in Unity 2022+
2. Load the main scene
3. Press play to start Unity scene
4. Adjust simulation parameters and click "Start Simulation"

Controls:

- Use "Q" and "E" to rotate the terrain left or right.
- Click anywhere on the terrain to ignite a fire.

Repository Structure:

...

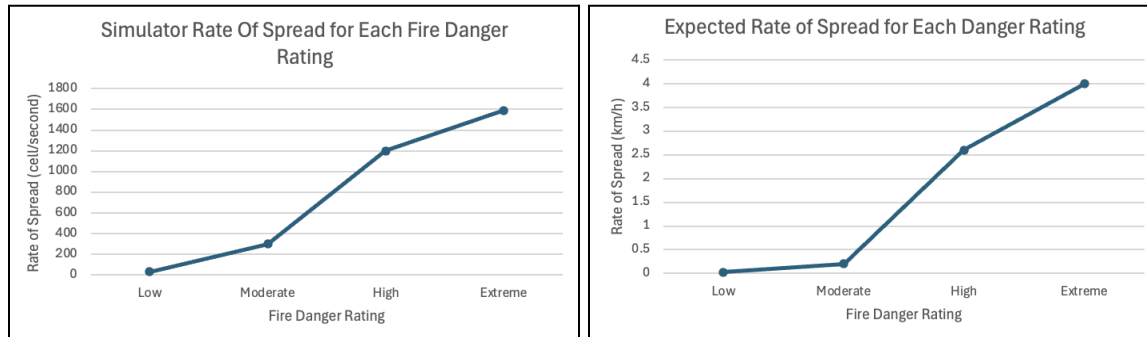
Assets/

- | | |
|-----------------|--|
| - Materials/ | *Unity Materials |
| - Prefabs/ | *Unity Prefabs |
| - Scenes/ | *Unity Scenes |
| - Scripts/ | *Simulation Logic |
| - Core/ | *Main logic controller |
| - Grid/ | *Grid creation and management |
| - SimMechanics/ | *Burning cell probability |
| - UI/ | *All user interaction management |
| - UnitTests/ | *unit tests, run through "Test Runner" |

...

c. APPENDIX C: ADJUSTING SIMULATOR RATE OF SPREAD

The adjusted rate of spread was determined by first mapping the expected and calculated rates of spread, which presented a very similar pattern to one another (Graphs 2 and 3 below). Then a ratio was obtained to determine how much the simulator sped up rate of spread in order to adjust the rate of spread values obtained during testing trials (Table 9 below).



Graphs 2 and 3: Fire Danger Rating mapped to Rate of Spread for simulator and expected conditions. Note the similarities in slope trends.

TABLE 9: Calculation of the value to adjust the simulator values to get a realistic estimate of rate of spread.

Fire Danger Rating	Max Expected Rate of Spread (km/h)	Average Simulator Rate of Spread (km/h)	Ratio of expected / simulator rate of spread
Low	0.05 km/h	30 km/h	600
Moderate	0.5 km/h	300 km/h	600
High	2 km/h	1230 km/h	615
Extreme	3 km/h	1590 km/h	530
Average Ratio →			586.25 → Round to 600